

Notes: Dempsey (RES, 2026)

Capital Requirements with Non-Bank Finance

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1 Big picture

- **Question.** What happens to aggregate activity, financial stability, and welfare when bank capital requirements rise in an economy where many firms can substitute from bank loans into non-bank finance?
- **Core friction.** Banks fund loans with insured deposits and costly equity, face limited liability, and can monitor borrowers; firms face moral hazard and choose between bank loans and direct non-bank borrowing when they have access to both.
- **Main result.** Tighter capital requirements reduce bank leverage and modestly reduce bank failures while having small effects on top-line output, consumption, capital, and welfare.
- **Mechanism.** Banks retain earnings and build net worth; non-bank competition disciplines bank loan pricing; corporate firms substitute across lenders, while bank-dependent non-corporates bear most costs.
- **Welfare.** The model's welfare optimum is slightly above the baseline requirement, around 9%.
- **Economic takeaway.** In a modern financial system with large non-bank financing options, capital regulation can make banks safer at low aggregate cost, but the costs are concentrated among firms without outside financing options.

2 Data and measurement

2.1 Debt composition and non-bank finance

Figure 1 documents the growth of non-bank debt sources for non-financial corporate business. The non-bank outside option in the model is a competitive corporate bond market, chosen because bonds are a major source of non-bank corporate finance and because they discipline bank pricing.

Interpretation: The figure motivates the central modelling choice: capital requirements should be evaluated in a credit system with lender substitution, not a bank-only credit system.

2.2 Firm types

The model has three firm types: A-rated corporates, B-rated corporates, and non-corporates. Non-corporate firms are fully bank-dependent. A-rated corporates are strong bond-market users; B-rated corporates can use bonds but face worse terms and stronger bank dependence.

Interpretation: The classification creates a laboratory in which substitution is strong for some firms and absent for others.

2.3 Bank variables and calibration targets

Banks are heterogeneous in net worth, choose loan prices and monitoring intensity, finance loans with net worth, deposits, and costly equity, and must satisfy a capital-to-risk-weighted-assets constraint. Tables 1 and 2 discipline the model using externally assigned parameters and internally calibrated target moments.

Interpretation: The calibration targets the margins through which the policy operates: bank balance sheets, debt composition, firm financing choices, and business-cycle credit patterns.

3 Models and quantitative specifications

3.1 Production and borrowing

Firms operate a risky decreasing-returns technology:

$$y = z\varepsilon(k^{\alpha_j}h^{1-\alpha_j})^{\eta_j}.$$

Capital combines firm endowment with borrowed funds:

$$k = \bar{k}_j + qb.$$

Interpretation: Firm borrowing is productive because it expands capital, but financing terms determine whether investment is feasible and how much agency risk remains.

3.2 Moral hazard and monitoring

After production, managers may divert a share of operating profits. Monitoring reduces the private gain from diversion:

$$\text{private diversion payoff} \propto x(1 - m)\theta_j\pi.$$

Default occurs when un-diverted operating profits are insufficient to repay debt.

Interpretation: Monitoring is the bank's technological advantage. It mitigates diversion and default risk, but it is costly and interacts with capital constraints and loan pricing.

3.3 Lender choice

A firm chooses among lenders by comparing contract value and a lender-specific preference shock:

$$\max_i V_{ij}(q_i, m_i; s) + \nu_i.$$

Interpretation: This structure generates smooth substitution across bank and non-bank finance when capital regulation changes bank contract values.

3.4 Bank balance sheet and capital requirement

Banks finance lending using net worth, insured deposits, and costly equity. The regulatory constraint requires

$$\frac{\text{equity}}{\text{risk-weighted assets}} \geq \chi.$$

Interpretation: The constraint reduces leverage and failure risk, but it may make banks reluctant to lend unless they build equity capital.

4 Identification and empirical design

This is a calibrated structural macro-finance model rather than a natural experiment. The credibility of the counterfactual comes from matching debt composition across lenders and firm types, bank balance-sheet moments, firm debt moments, business-cycle properties, and micro pricing patterns.

- The model must reproduce the split between bank and non-bank finance; otherwise the substitution channel is not credible.
- Bank leverage, dividends, net interest margins, chargeoffs, failure rates, and net worth discipline bank responses.
- Figure 3 validates the micro mechanism that weaker bank balance sheets push firms toward bonds and raise bank-bond spreads.
- Table 4 validates the model against cyclical macro, bank, and firm-financing facts.
- Table 7 shows robustness across model variants.
- The key limitation is that the non-bank sector is competitive bond finance, not a fully modelled levered shadow banking sector.

Interpretation: The paper’s empirical design is disciplined quantitative counterfactual analysis: the model is validated on multiple margins and then used to conduct policy experiments.

5 Tables and figures

5.1 Figure 1 — Growth of alternative debt sources for non-financial business in the U.S.

Read: The figure shows the expansion of corporate bonds and other debt sources relative to bank loans from 1960 to 2020.

Interpretation: This motivates the non-bank outside option in the model and shows why bank-only analyses of capital requirements are incomplete.

Economic message: Capital requirements transmit through the entire credit system, not only through bank balance sheets.

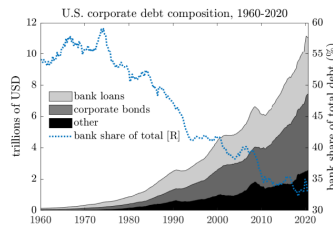


Figure 1: Growth of alternative debt sources for non-financial business in the U.S.

Notes: Figure is constructed using quarterly data from the Flow of Funds of the Financial Accounts of the United States, 1960Q1 through 2020Q4. Each shaded region represents the total volume of the specified type of debt for non-financial corporate firms. For details on the construction of each category, see Appendix B.

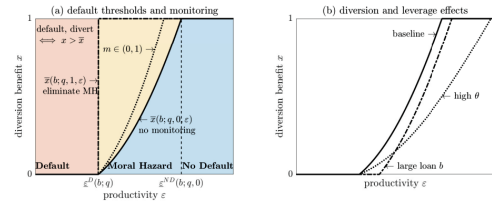


Figure 2: Default and diversion in firms' stage 3

Notes: Panel (a) plots the 3 regions in the interval $[0, \bar{x}]$ defined by the default and diversion cutoffs: (i) in the default region $[0, \bar{x}_D]$, $\bar{x} = 0$ and the manager (firm) always diverts (defaults); (ii) in the moral hazard region $[\bar{x}_D, \bar{x}_{ND}]$, the manager (firm) diverts (defaults) only if $x \geq \bar{x}$; and (iii) in the region $[\bar{x}_{ND}, \bar{x}]$ the manager (firm) never diverts (defaults). Panel (b) shows the

5.2 Figure 2 — Default and diversion in firms' stage 3

Read: The figure shows repayment, diversion, and default regions in the firm's state space.

Interpretation: It illustrates the moral-hazard region that bank monitoring can shrink.

Economic message: Bank monitoring has social value, but its macro importance depends on the size of the moral-hazard region.

5.3 Table 1 — Externally assigned parameters

Read: Table 1 lists preference, tax, loss-rate, firm-share, and regulatory parameters assigned outside the internal calibration.

Interpretation: External discipline reduces the degrees of freedom in the policy experiment.

Economic message: The model is tied to observed institutional and empirical features before counterfactual analysis.

Parameter		Value	Notes / Sources
Panel A: Household	discount factor	β	0.998 ann. risk free rate of 1%
	coef. of relative risk aversion	γ	2 standard
	labor disutility	A	2.5 standard
	loss rate in default	κ_d^B	0.67 Schwert (2020), Table 1
	loan issuance cost	κ_L	0.0029 1% annualized
Panel B: Firms	monitoring intensity	$\overline{m}_N$	0 normalization
	depreciation rate	δ	0.024 standard
	non-corporate firm share	$\tilde{\mu}_n$	0.304 Dyrdal and Pugsley (2019)
	A-rated share of corporates	$\frac{\tilde{\mu}_n}{1-\tilde{\mu}_n}$	0.44 Kisgen (2006)
Panel C: Banks	initial bank net worth	n_0	0 no initial net worth
	loss rate in default	κ_d^B	0.17 Schwert (2020), Table 1
	bank failure cost	κ_F	0.332 Elenev et al. (2020)
Panel D: Government	capital requirement	χ	0.08 Basel III
	gov't spending / GDP	$\frac{G}{Y}$	0.176 Elenev et al. (2020)
	labor income tax rate	τ_n	0.293 Elenev et al. (2020)
	corporate profit tax rate	τ_π	0.200 Elenev et al. (2020)
	interest income tax rate	τ_d	0.0315 Elenev et al. (2020), annualized
	deposit insurance fee rate	κ_d	0.00084 Elenev et al. (2020)
Panel E: Aggregates	average TFP	$\bar{z}$	1 normalization
	share of recession periods	$\bar{Q}_R$	0.211 share in postwar U.S.
	prob. exit recession	$Q_{RR'}$	0.167 avg. duration 1.5 years

Panel A: Target moments			Panel B: Calibrated parameters		
Moment	Data	Model	Parameter	Value	
A.1 Macro					
Output volatility (%)	1.03	1.05	Corp returns to scale	η_c	0.57
Capital to output ratio	2.20	2.09	Non-Corp returns to scale	η_n	0.69
Investment to output ratio	0.17	0.22	A-Corp capital share	α_a	0.15
Business debt to GDP ratio	0.72	0.86	B-Corp capital share	α_b	0.16
Labor share of income	0.58	0.52	Capital share, Non-Corp	α_n	0.14
A.2 Banks					
Bank leverage	0.92	0.92	A-Corp endowment	$\bar{k}_a$	0.91
Gross equity iss. rate (%)	1.10	0.32	B-Corp endowment	$\bar{k}_b$	0.61
Net dividend payout rate (%)	5.80	0.34	Non-Corp endowment	$\bar{k}_n$	0.90
C&I net interest margin (ann. %)	3.26	4.26	Corp SD productivity	σ_{sc}	0.20
C&I charge-off rate (%)	0.65	0.08	Non-Corp SD productivity	σ_{sn}	0.09
Bank failure rate (quarterly %)	0.74	0.79	Diversion benefit	ω	0.14
A.3 Firms					
Corporate leverage	0.41	0.41	Corp diversion rate	θ_c	0.83
Non-Corporate leverage	0.42	0.42	Non-Corp diversion rate	θ_n	0.42
A-Corp / B-Corp leverage ratio	1.26	1.40	Extreme value scale	ζ	0.0016
Corp bank debt / Total Corp debt	0.34	0.38	Recession TFP	z_R	0.984
Corp bank debt / Total bank debt	0.38	0.41	Equity issuance curvature	$\bar{\psi}$	0.08
Total Corp debt / Total debt	0.62	0.64	Mean entry cost	$\bar{\phi}_e$	0.18
Total Corp capital / Total capital	0.63	0.65	SD bank profit shocks	σ_{ϕ_p}	0.0087
A-Corp bond spread (ann. %)	1.62	1.74	Monitoring cost scale	$\bar{c}$	0.05

5.4 Table 2 — Target moments and internally calibrated parameters

Read: Table 2 lists moments and parameters chosen to match macro, bank, and firm financing facts.

Interpretation: This is the model's central calibration discipline.

Economic message: The counterfactual is credible because the calibrated model is forced to match the margins through which capital requirements operate.

5.5 Table 3 — Experiments to highlight key model features

Read: Table 3 defines variants altering moral hazard, monitoring costs, equity costs, and competition.

Interpretation: These variants identify which frictions matter for the baseline economy.

Economic message: The paper probes the channels behind the main policy result.

5.6 Figure 3 — Financing choices conditional on net worth of paired bank

Read: Firms matched with lower-net-worth banks use less bank debt and face higher bank-bond spreads.

<i>pct. chg. rel. to baseline parameterization</i>	Low MH [1]	Cheap Monitor [2]	Cheap Equity [3]	Easy Subs [4]	Cheap Entry [5]
Panel A: Macro					
Total output	-0.1	0.0	0.0	0.0	0.0
Total consumption	0.0	0.0	0.0	0.0	0.0
Total capital	0.2	0.0	0.1	0.1	0.0
Panel B: Banks					
Total bank lending	5.9	2.2	-0.3	1.7	-0.3
Total dividends	4.4	2.9	0.8	-0.3	2.2
Mass of active banks	1.0	-1.3	3.6	0.4	1.4
Average bank net worth	5.1	2.2	-0.2	1.5	0.1
C&I net interest margin	-0.8	-1.4	0.3	-1.5	0.2
C&I chargeoff rate	7.3	-5.3	-1.3	2.6	-0.5
Bank failure rate	-1.3	0.8	-3.3	-0.4	3.1
Panel C: Firms					
Total debt	0.6	0.0	0.2	0.1	0.0
Total Non-Corporate debt	0.8	0.4	0.1	0.1	0.0
Total Corporate debt	0.2	-0.6	0.6	0.2	0.1
Corp bank debt / Total Corp debt	13.3	5.9	-1.6	3.7	-0.9
Corp bank debt / Total bank debt	7.9	4.0	-1.3	2.1	-0.6
Total Corp debt / Total debt	0.2	0.3	-0.2	0.0	0.0

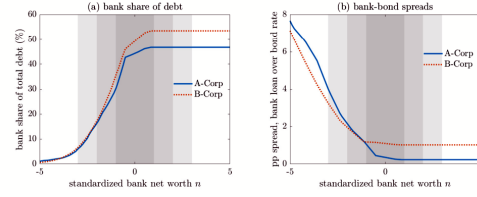


Figure 3: Financing choices conditional on net worth of paired bank

Notes: Each panel plots the indicated object against the standardized value of the net worth (under the average equilibrium distribution) of the firm's matched bank. For panel (b), the "bank-bond spread" is the difference between the (loan-weighted) loan rate charged by a bank with the indicated level of net worth and the equilibrium bond rate for the indicated firm type.

Interpretation: This validates the bank capital constraint channel at the micro level.

Economic message: Lender substitution prevents capital regulation from mechanically becoming a large aggregate credit contraction.

5.7 Table 4 — Business cycle properties: model vs data

Read: Table 4 compares volatility, cyclical, and persistence of macro, bank, and firm variables.

Interpretation: It validates the model beyond steady-state averages.

Economic message: The calibrated economy behaves plausibly over the cycle, which matters for macro-prudential policy.

	Data			Model		
	σ_x/σ_y [1]	ρ_{xy} [2]	$\rho_{xx'}$ [3]	σ_x/σ_y [4]	ρ_{xy} [5]	$\rho_{xx'}$ [6]
Panel A: Macro						
Total output	1.03%	1.0	0.9	1.05%	1.0	0.3
Total consumption	0.8	0.9	0.9	0.4	0.2	0.9
Total investment	5.2	0.9	0.9	19.3	0.9	0.2
Panel B: Banks						
C&I net interest margin	3.2	0.0	0.7	3.5	0.7	0.3
C&I chargeoff rate	43.4	-0.5	0.7	2.9	-0.7	0.2
Bank failure rate	0.4	-0.1	0.6	3.1	-0.2	0.9
Panel C: Firms						
Total debt	2.7	0.3	1.0	1.6	0.3	0.7
Non-Corporate debt	2.9	0.3	1.0	3.2	0.0	0.7
Corporate debt	2.8	0.4	0.9	1.2	0.7	0.6
Corporate bank debt	3.9	0.4	0.9	4.8	-0.5	0.5
Corporate bond debt	2.3	-0.2	0.9	3.7	0.7	0.3
Corp bank debt / Total Corp debt	1.6	0.4	0.9	5.1	-0.6	0.4
A-Corp bond spread	20.0	-0.4	0.8	6.1	0.3	-0.1
B-Corp bond spread	16.9	-0.5	0.8	3.2	0.3	-0.2

Table 4: Business cycle properties: model vs data

Notes: Columns [1] - [3] present the volatility relative to GDP (σ_x/σ_y), correlation with GDP (ρ_{xy}), and autocorrelation ($\rho_{xx'}$) of the indicated (logged, HP-filtered) moment for the model; columns [4] - [6] present the analogs for the data (see

<i>pct chg. rel. to 8% CR for CR =</i>	1%	5%	10%	15%	20%
	[1]	[2]	[3]	[4]	[5]
Panel A: Macro					
Total output	0.2	0.0	0.0	0.1	0.1
Total consumption	0.0	0.0	0.0	-0.1	-0.1
Total capital	0.4	0.1	0.1	0.1	0.2
Welfare (consumption equiv, pp)	-0.2	-0.1	0.0	-0.2	-0.2
Panel B: Banks					
Sectoral leverage	7.1	3.1	-2.1	-7.4	-12.7
Total bank lending	-0.5	0.0	1.2	-3.3	1.5
Total dividends	-18.7	-0.8	-3.7	-17.0	-23.2
Mass of active banks	23.2	2.3	4.3	17.3	29.1
Average bank net worth	-68.8	-30.2	21.5	66.5	126.0
Average monitoring intensity	-38.6	-18.2	9.7	37.9	55.9
C&I net interest margin	7.8	3.5	-3.8	-8.2	-16.9
C&I chargeoff rate	-12.8	-7.4	4.9	19.4	34.6
Bank failure rate	-12.3	0.3	-4.9	-15.6	-23.9
Bank failures	8.0	2.6	-0.8	-1.0	-1.8
Panel C: Firms					
Total debt	1.0	0.2	0.2	0.2	0.6
Total Non-Corporate debt	-2.0	-1.5	1.1	6.5	7.8
Total Corporate debt	6.5	3.4	-1.5	-11.1	-12.4
Corp bank debt / Total Corp debt	-8.3	-3.5	4.1	1.8	13.0

5.8 Table 5 — Long-run effects of capital requirement changes

Read: Table 5 shows small aggregate effects but large changes in bank leverage, net worth, monitoring, and sectoral composition.

Interpretation: This is the paper's central quantitative table: bank balance sheets change strongly, while top-line aggregates move little.

Economic message: Higher capital requirements can improve safety at low aggregate cost, but incidence is sectoral.

5.9 Figure 4 — Changes in select bank policies and distribution from 8% CR to 15% CR

Read: Figure 4 decomposes policy changes from the distributional shift in bank net worth.

Interpretation: The rightward shift in bank net worth offsets part of the direct lending contraction.

Economic message: Equilibrium bank capital accumulation is central to the muted aggregate cost.

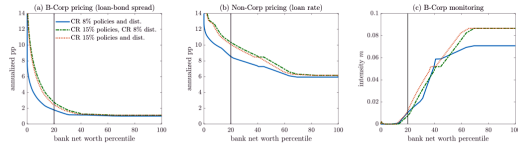


Figure 4: Changes in select bank policies and distribution from 8% CR to 15% CR

Notes: In each figure, the blue solid (red dotted) line corresponds to the banks' firm-type-specific policy indicated in the title given the percentile of net worth in the equilibrium bank distribution under a capital requirement of 8% (15%). The green dashed lines plot the 8% capital requirement policies translated onto the 15% distribution of bank net worth. The vertical line at the 20th percentile is an indicator to aid the discussion in the main text.

<i>pct change, 15% CR relative to baseline 8% CR</i>									
	Total	By firm			By lender		By firm-lender match		
Firm type	A-C	B-C	Non	all	all	A-C	A-C	B-C	B-C
Lender type	all	all	all	bank	bond	bank	bond	bank	bond
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Output	0.1	0.6	0.5	-1.0	0.0	0.2	3.9	-1.3	-0.8
Capital	0.1	2.9	2.4	-4.7	-1.1	1.9	6.8	0.7	1.3
Debt	0.2	8.7	5.0	-11.1	-3.3	5.5	14.4	5.5	4.2
Borrowing rate	-8.5	-24.6	-12.0	6.0	-3.4	-17.2	-27.8	-22.6	-9.0

Table 6: Detailed impacts of raising capital requirement to 15% by firm type

Notes: All moments are presented as the percent change in the indicated moment under a 15% capital requirement relative to the same moment under an 8% capital requirement. The loan rate is weighted by total debt in each case; for example, in columns [2] – [4] the loan rate is the loan-weighted average borrowing cost for each firm type, while in columns [5] – [6] the loan rate is the loan-weighted average borrowing cost at each lender, weighting across firm types. A-C and B-C refer to A- and B-rated corporates, respectively, while Non refers to non-corporates.

In order to understand how a higher capital requirement affects each firm type differently, Table

5.10 Table 6 — Detailed impacts of raising capital requirement to 15% by firm type

Read: Non-corporates experience the largest negative effects, while corporate firms are relatively insulated by substitution.

Interpretation: This table clarifies the incidence of regulation across firm types and lenders.

Economic message: The main cost of higher capital requirements is concentrated in bank-dependent borrowers.

5.11 Figure 5 — Response to negative bank net worth shock under 8% and 15% CR

Read: A 15% requirement dampens the fall in consumption, capital, debt, and bank activity after a bank net-worth shock.

Interpretation: The higher requirement raises resilience by reducing failure amplification.

Economic message: Capital requirements matter not only in steady state but also in crisis dynamics.

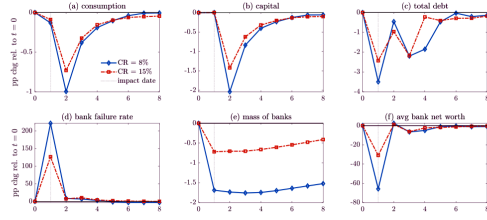


Figure 5: Response to negative bank net worth shock under 8% and 15% CR

Notes: In each figure, the blue solid (red dotted) line corresponds to the average response, relative to the date before

pct. chg. under 15% CR relative to 8% CR for indicated model variant	base [1]	low MH [2]	cheap monitor [3]	cheap equity [4]	easy subs [5]	cheap entry [6]
Panel A: Macro						
Total output	0.1	0.1	0.1	0.1	0.1	0.1
Total consumption	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Total capital	0.1	0.1	0.1	0.1	0.1	0.1
Welfare (cons. equiv. pps)	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Panel B: Banks						
Sectoral leverage	-7.4	-7.4	-7.4	-7.4	-7.4	-7.4
Total bank lending	-3.3	-2.4	-2.7	-3.1	-3.4	-3.6
Total dividends	-17.0	-14.9	-19.2	-18.9	-17.2	-16.9
Mass of active banks	17.3	17.1	20.9	19.8	17.5	17.3
Average bank net worth	66.5	68.9	67.4	67.1	66.4	65.5
Average monitoring intensity	37.9	40.0	9.9	36.7	33.6	38.3
C&I net interest margin	-8.2	-7.9	-8.3	-7.7	-8.3	-7.9
C&I chargeoff rate	19.4	16.9	24.1	18.9	20.2	19.0
Bank failure rate	-15.6	-15.8	-17.8	-16.9	-15.7	-15.6
Panel C: Firms						
Total debt	0.2	0.2	0.4	0.3	0.2	0.1
Total Non-Corporate debt	6.5	5.7	6.0	6.4	6.4	6.6
Total Corporate debt	-11.1	-9.6	-9.8	-10.5	-11.1	-11.4
Corp bank debt / Total Corp debt	1.8	1.2	1.4	1.8	1.2	1.5
Corp bank debt / Total bank debt	11.6	9.3	9.8	11.2	11.0	11.7
Total Corp debt / Total debt	6.2	5.4	5.5	6.0	6.2	6.4

Table 7: Effect of raising capital requirement from 8% to 15% across model variants

5.12 Table 7 — Effect of raising capital requirement from 8% to 15% across model variants

Read: Table 7 shows that the main policy result survives changes in moral hazard, monitoring, equity costs, and competition.

Interpretation: This is the robustness table for the paper’s central conclusion.

Economic message: The qualitative policy message is not driven by a single fragile calibration.

6 Short summary

- The paper studies capital requirements in a quantitative general-equilibrium model with heterogeneous firms, heterogeneous banks, and non-bank finance.
- Banks provide monitoring but are subject to insured-deposit risk shifting, costly equity issuance, and capital requirements.
- Corporate firms can substitute into bond finance; non-corporate firms are bank dependent.
- Tighter requirements reduce bank leverage and failure risk, but aggregate output, consumption, and capital move little.
- Banks build net worth by retaining earnings, which dampens the contractionary force of regulation.
- Non-bank competition disciplines loan pricing for corporate borrowers.
- The non-corporate sector bears most of the adverse effects.
- A higher requirement improves resilience to bank net-worth shocks.

7 Conclusion

7.1 Core conclusion

Dempsey shows that higher bank capital requirements can make banks safer with small aggregate macroeconomic costs in a modern credit system with non-bank finance.

7.2 Central mechanism

Banks build capital buffers; bond-market competition disciplines loan pricing; corporate firms substitute across lenders; bank-dependent firms bear the main costs.

7.3 Evidence chain

The paper motivates substitution with the growth of non-bank debt, calibrates the model to debt-composition, bank, firm, and business-cycle moments, and then uses counterfactuals to show modest aggregate effects and sharp sectoral incidence.

7.4 Policy implications

Capital regulation should be evaluated at the credit-system level. Aggregate effects can be small even when some sectors experience large credit contractions.

7.5 Caveats

The non-bank sector is modelled as competitive bond finance, not levered shadow banking, and the conclusions depend on the calibrated size of moral hazard and monitoring benefits.

7.6 Final takeaway

A capital requirement is not just a constraint on bank leverage. In equilibrium, it changes bank net worth, entry, lending terms, borrower substitution, and sectoral allocation. Those adjustments explain why tighter capital requirements reduce fragility at modest aggregate cost.